Evaluation of an Analytical Equivalent Hertzian Dipole Representation in TEM Cells Applying the Finite Element Method

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For the metrological quantification of radiated emissions of integrated circuits (ICs), the two-port transversal-electromagnetic (TEM) cell and IC stripline methods are utilized. Analytical formulas have been developed in the past to use the TEM cell measurement for the extraction of an equivalent analytical representation of the measured device under test (DUT). The equivalent sources are described as an array of electric and magnetic dipoles and can be used for, e.g., open-area emission test simulations. This article summarizes the theoretical background of this extraction methodology and gives an instruction for the needed measurements and calculations. Furthermore, the accuracy of this equivalent source model is evaluated against full-wave finite element simulations, and its portability to IC stripline measurements is investigated.

***Index Terms*— Dipole extraction method, electromagnetic compatibility, radiated emissions.**

I. Introduction

FINDING equivalent representations of the electromagnetic field originating from measured devices has been investigated for quite some time [1]. For radiated electromagnetic emissions (RE) of integrated circuits (ICs), three standardized measurement methodologies are available, one of them being the near-field scan method [2]. The focus of this article, however, lies on the two other methods, namely the transversal-electromagnetic (TEM) cell method and the IC stripline method, also standardized in [2]. For both methods, the device under test (DUT) is inserted into a TEM waveguide and the amplitude of the excited TEM wave is determined by measuring the output voltage of the cell with, e.g., an EMI receiver. The first formulations for an analytical extraction of an equivalent Hertzian dipole array to represent the DUT measured in a TEM cell were proposed by Sreenivasiah et al. [3] and have been adapted and further developed, e.g., in [4]. However, requirements for higher frequency ranges make it necessary to utilize the IC stripline method for RE measurements. This article investigates the portability of the analytic TEM model to the IC stripline method and its numerical verification by applying full wave finite element simulations. The rest of this article is organized as follows. Section II presents an analytical model for the description of the behavior of dipole sources inside TEM waveguides. Section III discusses the used finite element method (FEM) models of a generic TEM cell and an IC stripline. In Section IV, the equivalent source extraction is demonstrated for a simple test example. In addition, an experimental approach for a purely analytical emission estimation for a specific DUT is shown, which avoids the necessity of computational expensive numerical simulations. Section V will conclude the article.

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II. Theory of Dipole Moments in TEM Cells

For determining the RE of ICs the design of the IC itself plays an important role. It is reasonable that especially the bond wires might play a prominent role for unwanted RE when conducting high and fast switching currents. In [5] the relationship between current carrying paths inside cylindrical wave guides and the consequently excited EM waves is elaborated based on the Lorentz reciprocity theorem

$$\oiint_{\partial V} (\mathbf{E}_1 \times \mathbf{H_2} - \mathbf{E_2} \times \mathbf{H_1}) \cdot d\Gamma = \iiint_V (-\mathbf{E}_1 \cdot \mathbf{J_{e2}} + \mathbf{E_2} \cdot \mathbf{J_{e1}}) d\Omega \quad (1)$$

where $\mathbf{E}_1$ and $\mathbf{H}_1$ are the electric and magnetic field intensities due to the current density $\mathbf{J_{e1}}$ and $\mathbf{E}_2$ and $\mathbf{H}_2$ the intensities due to $\mathbf{J_{e2}}$ inside the waveguide volume V. This relationship is useful when deriving orthogonality properties of waveguide modes or computing the radiated field from currents inside waveguides. When following the elaborations, e.g., in [3] or [5] based on (1) a relationship between the output quantities of a TEM cell and the exciting Hertzian dipole moments inside the cell is found

$$\begin{pmatrix} a \\ b \end{pmatrix} = -\frac{1}{2}(\mathbf{m_e} \cdot \mathbf{e_0} \pm jk\mathbf{M_m} \cdot (\mathbf{x} \times \mathbf{e_0})) \quad (2)$$

with

$$\mathbf{e_0} = \left(\mathbf{z}e_{0z} + \mathbf{y}e_{0y}\right). \quad (3)$$

In (2), a and b are the amplitudes of the traveling waves in positive and negative x-direction inside the waveguide in $\sqrt{\mathrm{W}}$, $\mathbf{m_e}$ is the electric Hertzian dipole moment representing the DUT in Am, j is the imaginary unit, k is the wavenumber of the propagating mode in m^{-1}, $\mathbf{M_m}$ is the magnetic Hertzian dipole moment representing the DUT in Am^2, $\mathbf{x}$ is the unit vector in the propagation direction, $\mathbf{y}$ and $\mathbf{z}$ are the other unit vectors and $\mathbf{e_0}$ is the electrical field strength inside the

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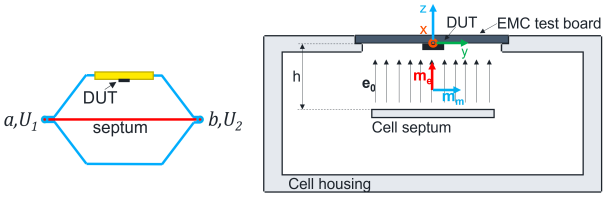

(a) (b)

Fig. 1. (a) Principle sketch of a TEM cell. (b) Cross-sectional view of the TEM cell with exciting dipole moments inside the cell volume.

waveguide, when excited with unit power and is therefore dependent only on the geometry of the waveguide as can be seen from Fig. 1(b).

Due to this normalization to unit power the dimension of $\mathbf{e_0}$ is $((V/m)/\sqrt{1W})$. Since a symmetrical waveguide is assumed and the equivalent sources are placed in the center of the guide concerning y, e_{0y} can be set to zero. The correlation of $\mathbf{e_0}$ to the geometry of a waveguide is described in detail in [6]. For this work, the simplification

$$e_{0z} = \frac{\sqrt{Z_W}}{h} \tag{4}$$

based on [3] is used, where Z_W is the wave impedance of the waveguide and h is the distance from the septum of the guide to the housing in the z-direction [see Fig. 1(b)]. Due to the normalization to 1 W, the excitation power disappears in (4). Since the expression $\mathbf{M_m}$ of dimension Am^2 is not used by the utilized finite element code and by most literature, it will for this work be substituted by the following equation:

$$\mathbf{m_m} = \mathbf{M_m}\mu_0\omega \tag{5}$$

In (5), the dimension of $\mathbf{m_m}$ is Vm, μ_0 is the magnetic field constant and "ω" is the angular frequency. With $k = \mu_0\omega/Z_0$, (2) can be rewritten as follows:

$$\begin{pmatrix} a \\ b \end{pmatrix} = -\frac{1}{2}e_{0z}\left(m_{ez} \mp j\frac{m_{my}}{Z_0}\right). \tag{6}$$

Z_0 is the free space wave impedance. Since it is more convenient to represent the output of the TEM cell as voltages instead of power waves, a and b are expressed through

$$\begin{pmatrix} U_1 \\ U_2 \end{pmatrix} = \begin{pmatrix} a \\ b \end{pmatrix}\sqrt{Z_W}. \tag{7}$$

Substituting (4) into (6) and subsequently (6) into (7) yields

$$\begin{pmatrix} U_1 \\ U_2 \end{pmatrix} = -\frac{Z_W}{2h}\left(m_{ez} \mp j\frac{m_{my}}{Z_0}\right). \tag{8}$$

Through addition, respectively, subtraction of (8) we obtain

$$U_1 + U_2 = -\frac{Z_W}{h}m_{ez} \tag{9}$$

and

$$U_1 - U_2 = j\frac{Z_W}{h}\frac{1}{Z_0}m_{my}. \tag{10}$$

Rearranging (9) and (10) gives the magnitudes of the two extracted dipole sources

$$|m_{ez}| = \frac{|U_1 + U_2|h}{Z_W}, \; |m_{my}| = \frac{|U_1 - U_2|hZ_0}{Z_W}. \tag{11}$$

The quantities $|U_1 + U_2|$ and $|U_1 - U_2|$ can be measured, e.g., with the help of a hybrid junction rat race coupler.

III. Finite Element Models

A. Idealized TEM Cell

To validate the presented analytical model, a generic TEM cell with $Z_W \approx 50\ \Omega$ was modeled in a commercial full-wave finite element simulation suite according to Fig. 2. This model

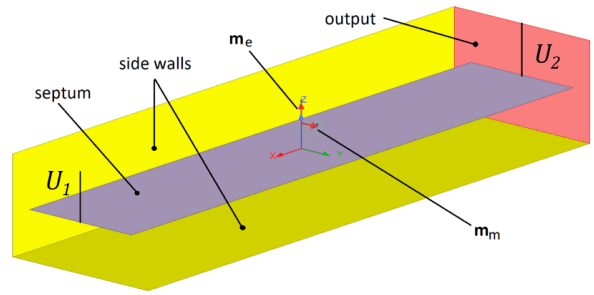

Fig. 2. Idealized TEM cell excited with an electric and a magnetic Hertzian dipole moment.

TABLE I

Output Voltages of the TEM Cell

Source active	Min. Voltage	Max. Voltage	Frequency Range
Ele. dipole, 3.2e-4 Am	0.990 V	0.991 V	10 MHz to 3 GHz
Mag. dipole, 0.1206 Vm	0.988 V	0.989 V	

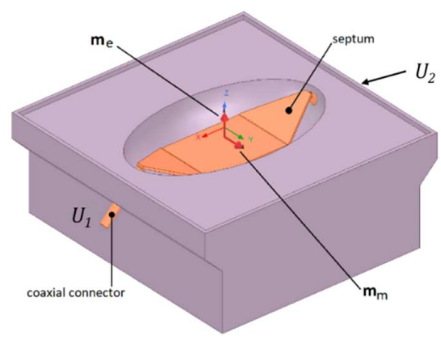

Fig. 3. IC stripline excited with an electric and a magnetic Hertzian dipole moment.

represents the exact idealized geometry which is considered for the analytical model. It consists of seven surfaces: four sidewalls of the guide (yellow), two outputs (red), and the septum (gray). For better visibility, two of the sidewalls and one output are faded out. The sidewall and septum are assigned with perfectly electric conducting boundary conditions. The outputs are assigned with wave ports to terminate the waveguide. The cross section of the guide is 16 × 26.592 mm. The septum lies exactly in the center and is 20 mm wide. The length of the simplified TEM cell is 100 mm.

The waveguide is excited with an electric Hertzian dipole moment $\mathbf{m_e}$ oriented in the z-direction and a magnetic Hertzian dipole moment $\mathbf{m_m}$ in the y-direction. Both are placed in the upper half of the guide in the center between the septum and upper wall. For a desired output voltage of the guide of 1 V (computed along the black lines) at the outputs in Fig. 1, the needed intensities according to (11) and the FEM-computed output voltages are given in Table I.

The deviations from one Volt can be explained mainly by the simplification (4).

B. IC Stripline

In the next step, the same experiment is done inside a more realistic geometry shown in Fig. 3. It shows a model of an IC stripline as described in [2]. The real-life parent of this model is used for EMC characterizations of ICs and sensors in the ams OSRAM laboratories. The DUT to septum distance is 8 mm as before and the stripline has a nominal impedance of 50 Ω. The material of the septum and the coaxial connectors is copper and the block is steel. On the top side, a 2 mm strong

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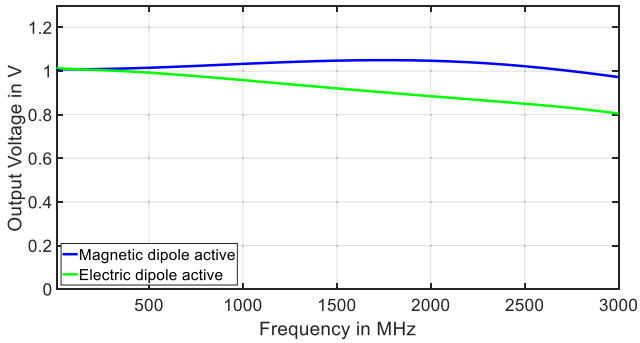

Fig. 4. Output voltages of the IC stripline when excited with the dipole intensities listed in Table I.

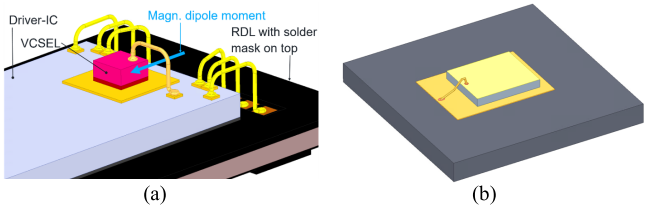

Fig. 5. (a) Rendered 3-D depiction of a functional VCSEL product. (b) Simplified model of a VCSEL device used as a test vehicle for this work.

lid of copper covers the stripline (faded out in Fig. 3). The coaxial connectors are also 50 Ω and terminated with wave ports in the FEM model. The output voltage is measured at the end of the connectors.

The geometry is excited with the same dipole array as for the generic TEM cell. The excitations are again modeled as incident dipole fields and are placed in the center between the septum and lid. The computed results of this simulation setup are shown in Fig. 4.

It can be seen, that for higher frequencies deviations from the expected value of 1 V can be observed. The reason for this behavior can be found in the fact that at higher frequencies not only the fundamental TEM mode occurs and higher-order modes start to be present. The computed deviations from 1 V can also be used as a correction factor for the presented dipole extraction methodology for this specific IC stripline geometry, as will be shown in Section IV.

IV. Test Example

Since the applicability of the TEM cell-based theory to the IC stripline method could be shown numerically, the desired dipole extraction methodology should be tested on a realistic problem.

A. Model and Dipole Extraction

The presented analytical extraction methodology is demonstrated by means of a simple test structure. This test structure represents a very simplified version of a vertical cavity surface emitting laser (VCSEL) device in a stacked die configuration. It consists of a lower die, which represents the driver IC of a VCSEL device and an upper die representing the passive, low-impedance material VCSEL die itself. Both dies are connected by the lower side of the VCSEL (cathode) and a JEDEC [7] gold bond wire to the upper side of the VCSEL (anode). Fig. 5(a) shows a rendering of a real-life VCSEL product and Fig. 5(b) shows the simplified model used for these investigations. This type of structure is chosen as a test device because high RE is expected for such devices due to the high operating currents required through the loop-shaped bond wire.

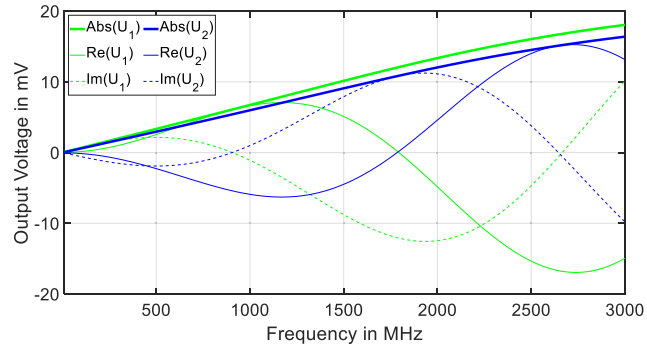

Fig. 6. Output voltages of the IC stripline for the RE simulation of the test device.

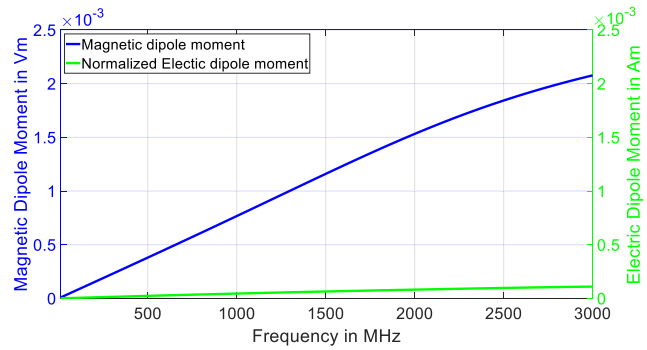

Fig. 7. Extracted Hertzian dipole moments of the test device calculated from the results in Fig. 6 derived from (11).

For the dipole extraction, the simplified VCSEL device is inserted upside down into the IC stripline in an orientation, where the bond wire is parallel to the septum. This way, the highest coupling and hence the highest RE are expected. In the combined simulation model of IC stripline and test device, the bond wire current path is excited with a constant current of 1 A in the frequency range between 10 MHz and 3 GHz. The output voltages at the IC stripline ports are computed and the electric and magnetic dipole moments are calculated according to (11). Fig. 6 shows the results for the computed voltages at both ports of the IC stripline and Fig. 7 shows the calculated dipole moments. In Fig. 7, the electric dipole moment is normalized by the value of the free space wave impedance of 377 Ω. In this way, it is better comparable to the magnetic moment in terms of how they contribute to the output voltage of the IC stripline, according to (8).

As can clearly be seen from Fig. 7 and as expected, the electromagnetic behavior in this orientation is dominated by the magnetic dipole moment.

B. Numerical Test Using FEM

Now the test device is removed from the FEM model of the IC stripline and replaced by the extracted dipole moments of Section IV-A, which in the following is referred to as backpropagation (BP). The extracted dipole moments are multiplied with the correction factors of Fig. 4, to account for the non-ideal behavior of the IC stripline in the higher frequency range. In other words, the same simulation model as depicted in Fig. 3 is used, but with the intensity of the two dipole moments set accordingly. Again, the voltages U_1 and U_2 at the coaxial terminals are computed and are shown in Fig. 8. As one can see, due to the now given symmetry the magnitude of the voltages U_1 and U_2 perfectly agree, which was not the case for the test device. The computed magnitudes give exactly

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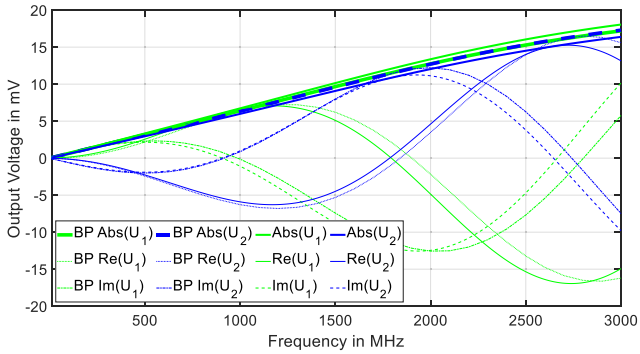

Fig. 8. Results for the IC stripline output voltages of the FEM BP with the extracted Hertzian dipole moments compared to the original result of the test device simulation.

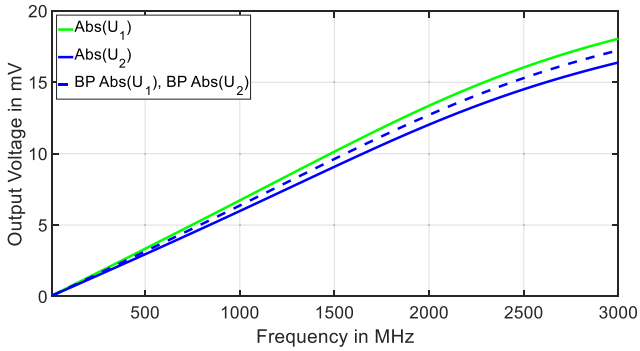

Fig. 9. Results for the absolute values of the IC stripline output voltages of the analytical BP with the extracted Hertzian dipole moments compared to the original result of the test device simulation.

Fig. 10. Side view of the test device with the surface area of the bond wire current path.

the average value of the two test device values. This shows, that the analytical model cannot represent different levels of the two output voltages [as can also be seen in (8)].

Hence, the extraction methodology is a good approximation of the real test device but cannot represent all influencing effects of the die and the bond wire.

C. BP Analytical

With the knowledge of the equivalent dipole array, also an analytical calculation of the IC stripline output voltages can be done using (8). For this BP, the geometry correction factors of Fig. 4 have to be used again. This method gives the exact same results as the FEM BP as can be seen in Fig. 9, while at the same time avoids the FE modeling of the DUT completely.

Note that since all wave terms are omitted when deriving (8), only the absolute values of the IC stripline output voltages can be calculated analytically.

D. Analytical Dipole Extraction

In the next step, an approach is made to do a purely analytical estimation for the RE emission prediction of our test device. For this, we use as naive first order estimations for the dipole moments

$$|m_{ez}| = 0 \tag{12}$$

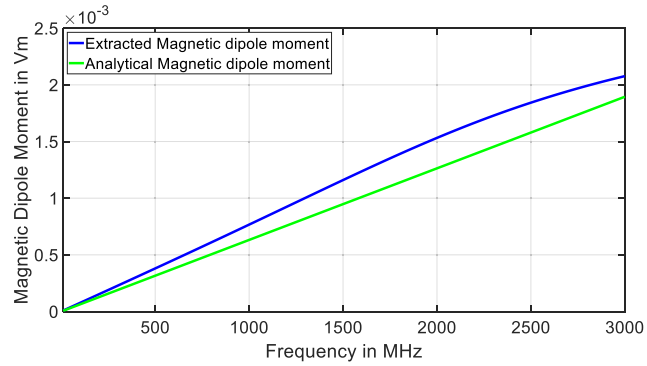

Fig. 11. Comparison between the extracted magnetic dipole moment and the purely analytical estimation applying (13).

and

$$|m_{my}| = |I_m l| = |j\Gamma\omega\mu_0 I_0|. \tag{13}$$

Equation (13) is based on [8] where Γ is the surface area covering the current path, as shown in Fig. 10. Fig. 11 finally shows the result of this approach for the magnetic dipole moment in comparison with that obtained from Fig. 7.

As can be seen, this simple analytical approach already gives a good approximation for the dipole intensities and subsequently for the pre-estimation of the radiated emissions of our simplified VCSEL test device. Since numerical simulations are completely avoided in this approach, the computational effort is reduced by magnitudes compared to FEM.

V. Conclusion

In this article, we show the applicability of analytic dipole representations originating from TEM cell test methodologies for IC stripline tests (used for the EMC characterization of ICs). For conformation, the applicability was investigated by means of a simple test device.

Moreover, a purely analytical method for the pre-estimation of RE of devices with high current conducted on bond wires is presented to avoid computational expensive FEM simulations.

In future work, this method will be refined and extended to SMD devices like capacitors, which are an accompanying component of VCSEL-based applications, too.

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